

Miniproject 2 - Creating a ZCB Term Structure

FRE 6103 Valuation for Financial Engineering, NYU Tandon | Raj Pawar | Methodology

Data and conventions. Wall Street Journal / Tullett Prebon 3 pm quotes for all 353 US Treasury notes and bonds on Sep 4, 2026 (the Week 2 data file). Quotes are clean prices in 32nds with a third decimal in eighths of a 32nd ($99.256 = 99 + 25.75/32$); we price off the asked side. Regular-way settlement is T+1 business days, Sep 8, 2026 (Monday Sep 7 is Labor Day). Both conventions were verified by recomputing the WSJ asked yield of every bond: median discrepancy 0.02 bp, RMSE 0.5 bp, versus 1.8 bp for the next-best settlement rule and over 86 bp if the quotes are read as decimals.

Cash flows. Each bond pays coupon/2 semi-annually on its maturity day-of-month (month-end maturities pay on month-ends) and 100 at maturity. Accrued interest is actual/actual within the coupon period, coupon/2 x days accrued / days in period, and dirty price = clean price + accrued (slide 34). Time to each payment is $t = \text{days}/365$ from settlement. Street yields use the Treasury convention (semi-annual compounding with a fractional first period, simple interest when one payment remains), which reproduces Excel's YIELD.

Model. The continuously compounded zero rate follows the Svensson (1994) form suggested in the data file, $r(t) = b_0 + b_1 (1 - e^{-(t/T_1)})/(t/T_1) + b_2 [(1 - e^{-(t/T_1)})/(t/T_1) - e^{-(t/T_1)}] + b_3 [(1 - e^{-(t/T_2)})/(t/T_2) - e^{-(t/T_2)}]$. The discount factor is $d(t) = e^{-r(t)t}$ and each bond's model dirty price is the sumproduct of its payment stream with the discount factors, $\sum_k CF_k d(t_k)$ (slide 33). The form nests Nelson-Siegel ($b_3 = 0$), is smooth, has finite limits at both ends ($r(0) = b_0 + b_1$, $r(\infty) = b_0$) and gives the instantaneous forward curve in closed form, $f(t) = b_0 + b_1 e^{-(t/T_1)} + b_2 (t/T_1) e^{-(t/T_1)} + b_3 (t/T_2) e^{-(t/T_2)}$.

Estimation. The six parameters minimise $\sum_i [(P_i(\text{model}) - P_i(\text{market})) / (P_i(\text{market}) D_i)]^2$, each price error divided by dirty price times modified duration, which is the first-order yield error, so the objective is the sum of squared yield errors (slide 31) while each model price stays a linear sumproduct. Bonds with under 3 months to maturity (14 of 353) are excluded from estimation, as in Gurkaynak, Sack and Wright (2007): their prices reflect money-market conditions and, with near-zero duration, a one-cent quote error is a 50 bp yield error. They are still priced off the curve as an out-of-sample check. Constraints: $b_0 \geq 0$, $T_1, T_2 \geq 0.05$. Because the objective is multimodal in (T_1, T_2) , the least-squares solver is run from 19 starting points on a tau grid; 11 of them reach the same optimum. In Excel the identical objective is the sum-of-squares cell on the Curve sheet, minimised with Solver (GRG Nonlinear); the workbook reproduces every Python model price, yield and zero rate.

Results. $b_0 = 0.0000$ (non-negativity bound active), $b_1 = 0.0363$, $b_2 = 0.0447$, $b_3 = 0.1622$, $T_1 = 1.76$, $T_2 = 15.93$. The curve prices the 339 bonds with a yield RMSE of 2.9 bp (mean absolute 2.1 bp; worst 14.8 bp for the 4.375% 2026-12-15 bond). Nelson-Siegel on the same data gives 7.5 bp, so the second hump is needed. The zero curve rises from 3.63% at the short end to 4.14% at 1 year, 4.49% at 5, 4.79% at 10, 5.33% at 20 and 5.31% at 30 years, peaking at 5.38% around 24 years. The deliverable table lists $r(t)$, $d(t)$ and $f(t)$ for all 245 distinct Treasury payment dates from Sep 15, 2026 to Aug 15, 2056. The asymptotic level b_0 is weakly identified by a 30-year sample: pinning it anywhere from 0% to 3% worsens the fit by only 0.3 bp RMSE and moves the in-sample curve by about 10 bp at the two ends (a 3% change in $d(30y)$), so the payment-date discount factors are robust to it; with $b_0 = 0$ the fitted forward rate decays beyond the sample, so the curve should not be extrapolated past 30 years.